# System and Devices (SYS 3)

## Lecture 12: Superscalar Processors

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COM00018I

# Simple In-Order Pipelining

- Pipelining – Goal was to complete one instruction per clock cycle.

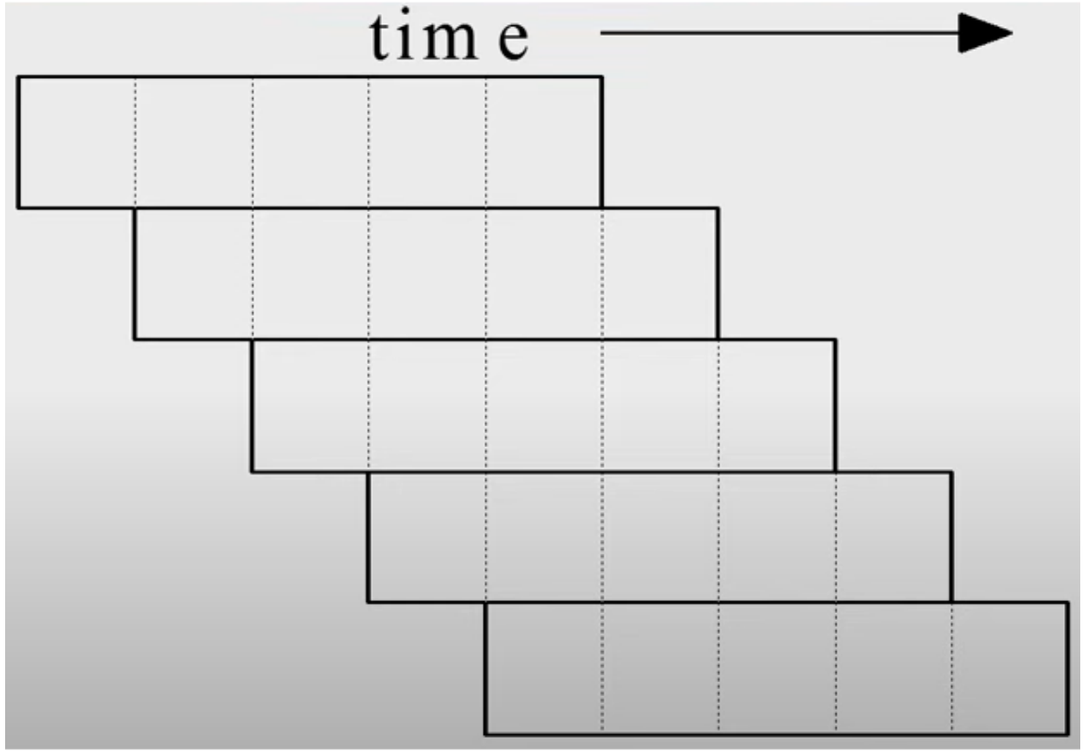
Instruction 1

Instruction 2

Instruction 3

Instruction 4

Instruction 5



# Advanced Pipelining

- Increase the depth of the pipeline to increase the clock rate – **Superpipelining**
- Fetch (and execute) more than one instructions at one time (expand every pipeline stage to accommodate multiple instructions) – **Multiple-issue (superscalar)**
- Launching multiple instructions per stage allows the instruction execution rate, CPI, to be less than 1

# Superpipelining

- Superpipelining – Increase the depth of the pipeline to increase the clock

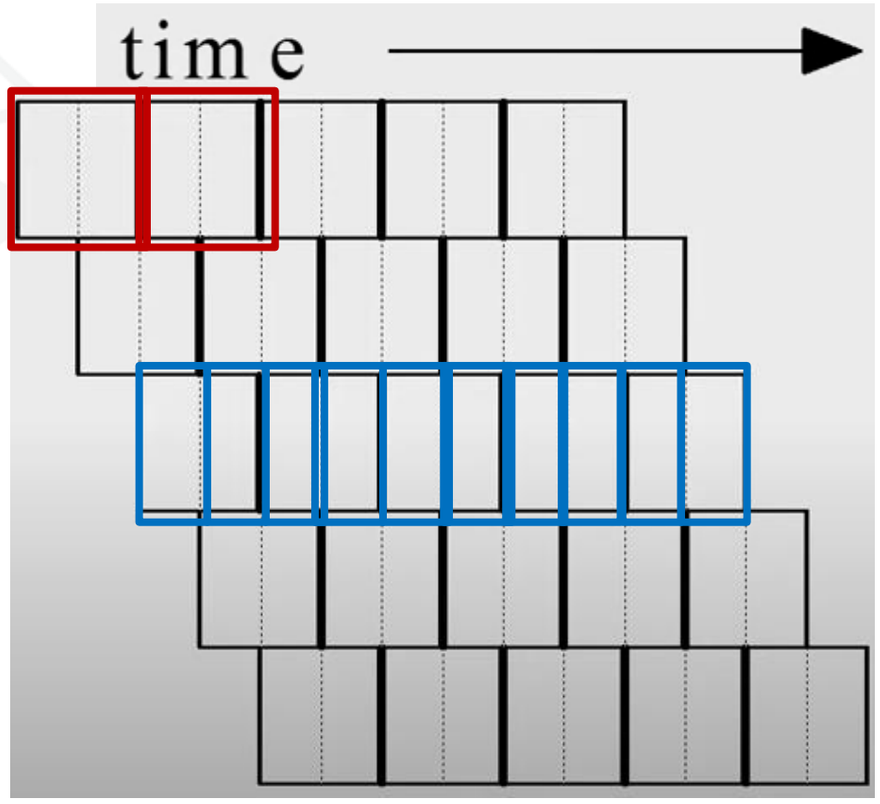
Instruction 1

Instruction 2

Instruction 3

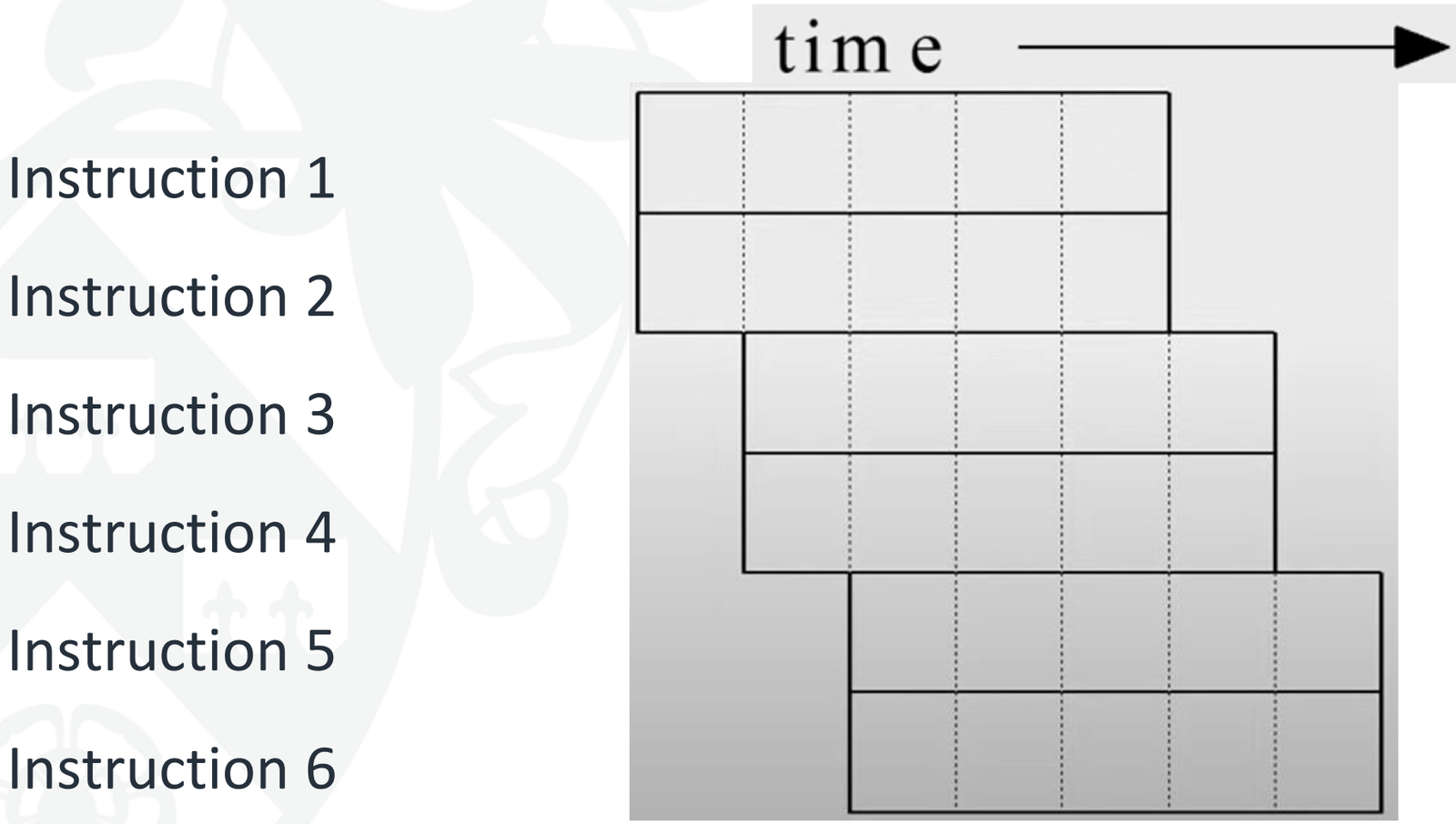
Instruction 4

Instruction 5



# Superscalar – Multiple-Issue

- Fetch (and execute) more than 1 instructions at one time
- Expand every stage to accommodate multiple instructions



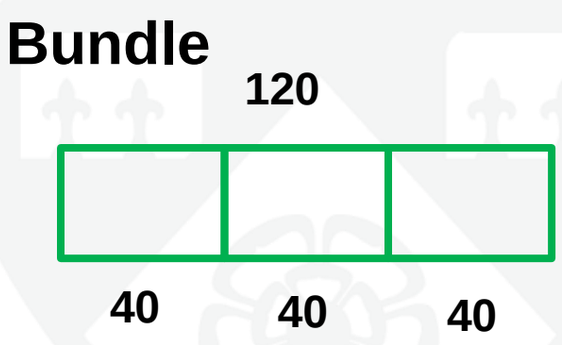
# Multiple-Issue and Static Scheduling

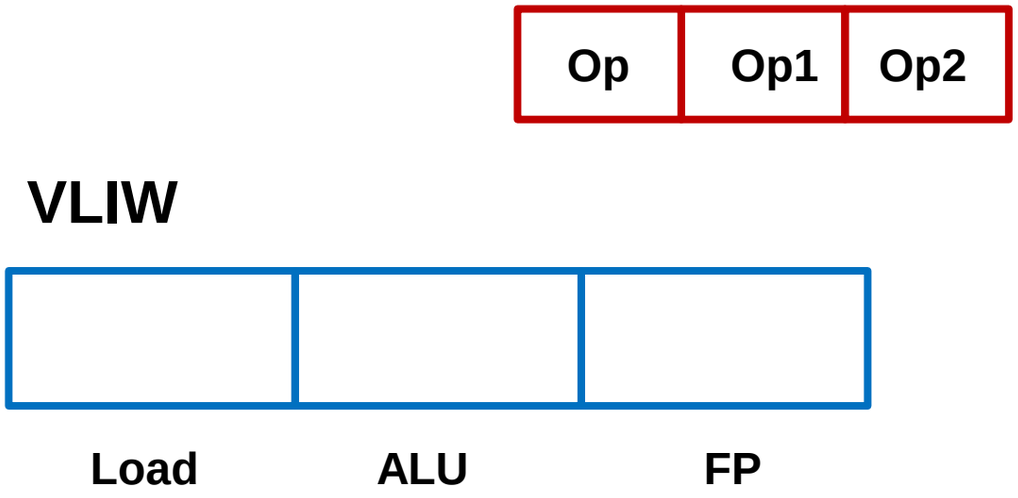
- Basic pipeline will give only a min CPI of 1
- To achieve CPI < 1, need to complete multiple instructions per clock (**Superscalar processors**)
- Multiple functional units and Multiple Issue
- Solutions:

  - Statically scheduled superscalar processors
  - VLIW (very long instruction word) processors
  - Dynamically scheduled superscalar processors

# VLIW Processors

- Package multiple operations into one instruction (**long word**)
- The instruction have a set of fields for each functional unit
- 16 to 24 bits per unit, → instruction length of 80 to 120 bits
- Must be enough parallelism in code to fill the available slots
- Find enough parallel instructions by loop unrolling and scheduling



# VLIW Processors

- Disadvantages:
  - Statically find parallelism
  - Lack of parallelism – slot waste
  - Code size increase due to unrolling
  - No hazard detection hardware

# VLIW Processors

```
Loop:   L.D      F0, 0(R1)      ; F0=array element
        ADD.D    F4, F0, F2     ; add scalar in F2
        S.D      F4, 0 (R1)     ; store result
        DADDUI R1, R1, #-8      ; decrement pointer
                                ; 8 bytes (per DW)
        BNE      R1, R2, Loop   ; branch R1!=R2
```

| | | | Clock cycle issued |
|---|---|---|---|
| Loop: | L.D | F0, 0(R1) | 1 |
| | stall | | 2 |
| | ADD.D | F4, F0, F2 | 3 |
| | stall | | 4 |
| | stall | | 5 |
| | S.D | F4, 0 (R1) | 6 |
| | DADDUI | R1, R1, #-8 | 7 |
| | stall | | 8 |
| | BNE | R1, R2, Loop | 9 |

- Example VLIW processor:
  - One integer instruction (or branch)
  - Two independent floating-point operations
  - Two independent memory references

**VLIW**

| I | M1 | M2 | FP1 | FP2 |
|---|---|---|---|---|

# VLIW Processors

- Example VLIW processor:
  - One integer instruction (or branch)
  - Two independent floating-point operations
  - Two independent memory references

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| | stall | | 8 |
| | BNE | R1, R2, Loop | 9 |

| | Memory reference 1 | Memory reference 2 | FP operation 1 | FP operation 2 | Integer operation/branch |
|---|---|---|---|---|---|
| 1 | L.D F0,0(R1) | L.D F6,-8(R1) | | | |
| 2 | L.D F10,-16(R1) | L.D F14,-24(R1) | | | |
| 3 | L.D F18,-32(R1) | L.D F22,-40(R1) | ADD.D F4,F0,F2 | ADD.D F8,F6,F2 | |
| 4 | L.D F26,-48(R1) | | ADD.D F12,F10,F2 | ADD.D F16,F14,F2 | |
| | | | ADD.D F20,F18,F2 | ADD.D F24,F22,F2 | |
| 5 | S.D F4,0(R1) | S.D F8,-8(R1) | ADD.D F28,F26,F2 | | |
| . | S.D F12,-16(R1) | S.D F16,-24(R1) | | | DADDUI R1,R1,#-56 |
| . | S.D F20,24(R1) | S.D F24,16(R1) | | | |
| . | S.D F28,8(R1) | | | | BNE R1,R2,Loop |

VLIW instructions that occupy the inner loop and replace the unrolled sequence.

Performance: 9 cycles to complete 26 instructions. More registers used.

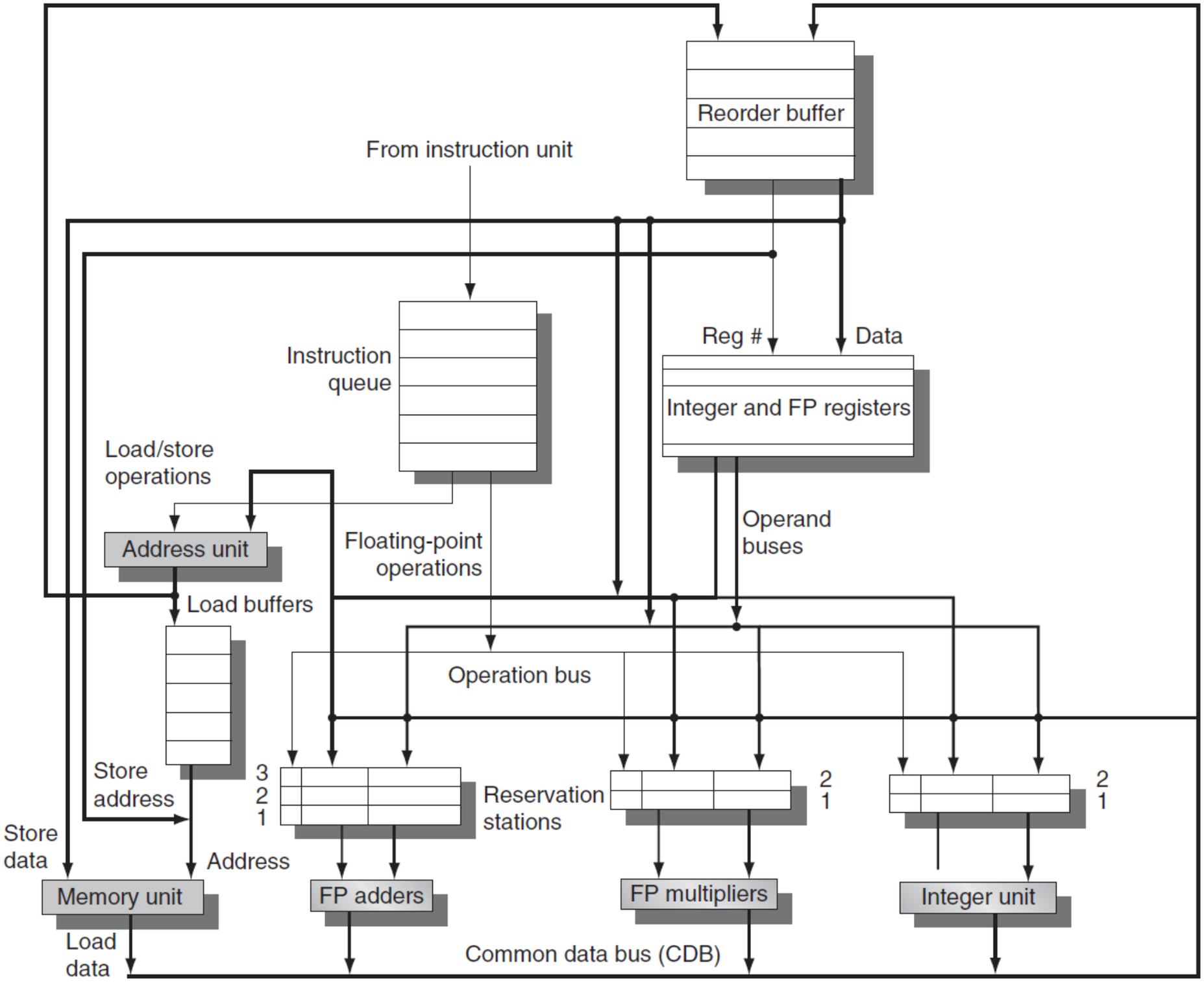
# Extreme Optimization

- Modern micro-architectures use this triple combination:
  - Dynamic scheduling + multiple issue + speculation
- Dependency between instructions issued in same clock.
- Register read for multiple instructions in parallel.
- Complex Issue logic to check dependencies and hazards.
- Assign reservation stations and ROB entries in a cycle.

# Handling Multiple Issue

- Assign RS and ROB for every instruction in next issue bundle.
- Limit the number of instructions of a given class that can be issued in a bundle [one FP, one integer, one load-store etc].
- Stall Issue if ROB not available.
- Examine all the dependencies among the instructions in the bundle.
- If dependencies exist in bundle, encode the in reservation stations with ROB entry.
- Multiple completion/commit (wide CDB+ROB update)

# Handling Multiple Issue




- One issue per unit per cycle
- Wider operand buses, CDB
- Multiple register read/cycle
- Multiple instruction to commit/cycle

# Example

```
Loop:   LD      R2, 0(R1)       ; R2=array element
        DADDIU  R2, R2, #1      ; increment R2
        SD      R2, 0(R1)       ; store result
        DADDIU  R1, R1, #8      ; increment pointer
        BNE     R2, R3, LOOP    ; branch if not last element
```

# Example (Without Speculation)

| Iteration number | Instructions | Issues at clock cycle number | Executes at clock cycle number | Memory access at clock cycle number | Write CDB at clock cycle number | Comment |
|---|---|---|---|---|---|---|
| 1 | LD R2,0(R1) | 1 | 2 | 3 | 4 | First issue |
| 1 | DADDIU R2,R2,#1 | 1 | 5 | | 6 | Wait for LW |
| 1 | SD R2,0(R1) | 2 | 3 | 7 | | Wait for DADDIU |
| 1 | DADDIU R1,R1,#8 | 2 | 3 | | 4 | Execute directly |
| 1 | BNE R2,R3,LOOP | 3 | 7 | | | Wait for DADDIU |
| 2 | LD R2,0(R1) | 4 | 8 | 9 | 10 | Wait for BNE |
| 2 | DADDIU R2,R2,#1 | 4 | 11 | | 12 | Wait for LW |
| 2 | SD R2,0(R1) | 5 | 9 | 13 | | Wait for DADDIU |
| 2 | DADDIU R1,R1,#8 | 5 | 8 | | 9 | Wait for BNE |
| 2 | BNE R2,R3,LOOP | 6 | 13 | | | Wait for DADDIU |
| 3 | LD R2,0(R1) | 7 | 14 | 15 | 16 | Wait for BNE |
| 3 | DADDIU R2,R2,#1 | 7 | 17 | | 18 | Wait for LW |
| 3 | SD R2,0(R1) | 8 | 15 | 19 | | Wait for DADDIU |
| 3 | DADDIU R1,R1,#8 | 8 | 14 | | 15 | Wait for BNE |
| 3 | BNE R2,R3,LOOP | 9 | 19 | | | Wait for DADDIU |

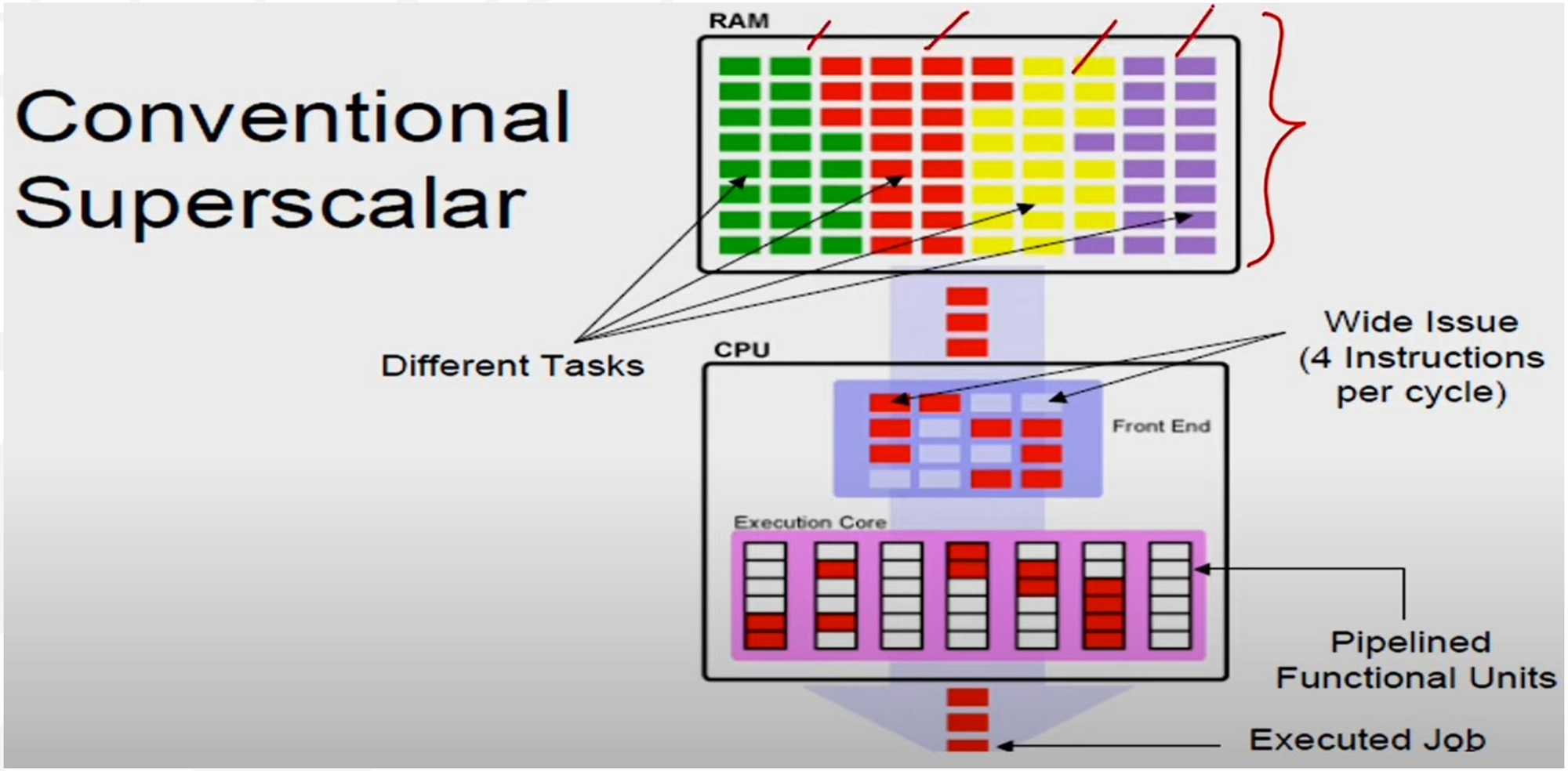
**The time of issue, execution, and writing result for a dual-issue version of our pipeline *without* speculation.**

# Example (With Speculation)

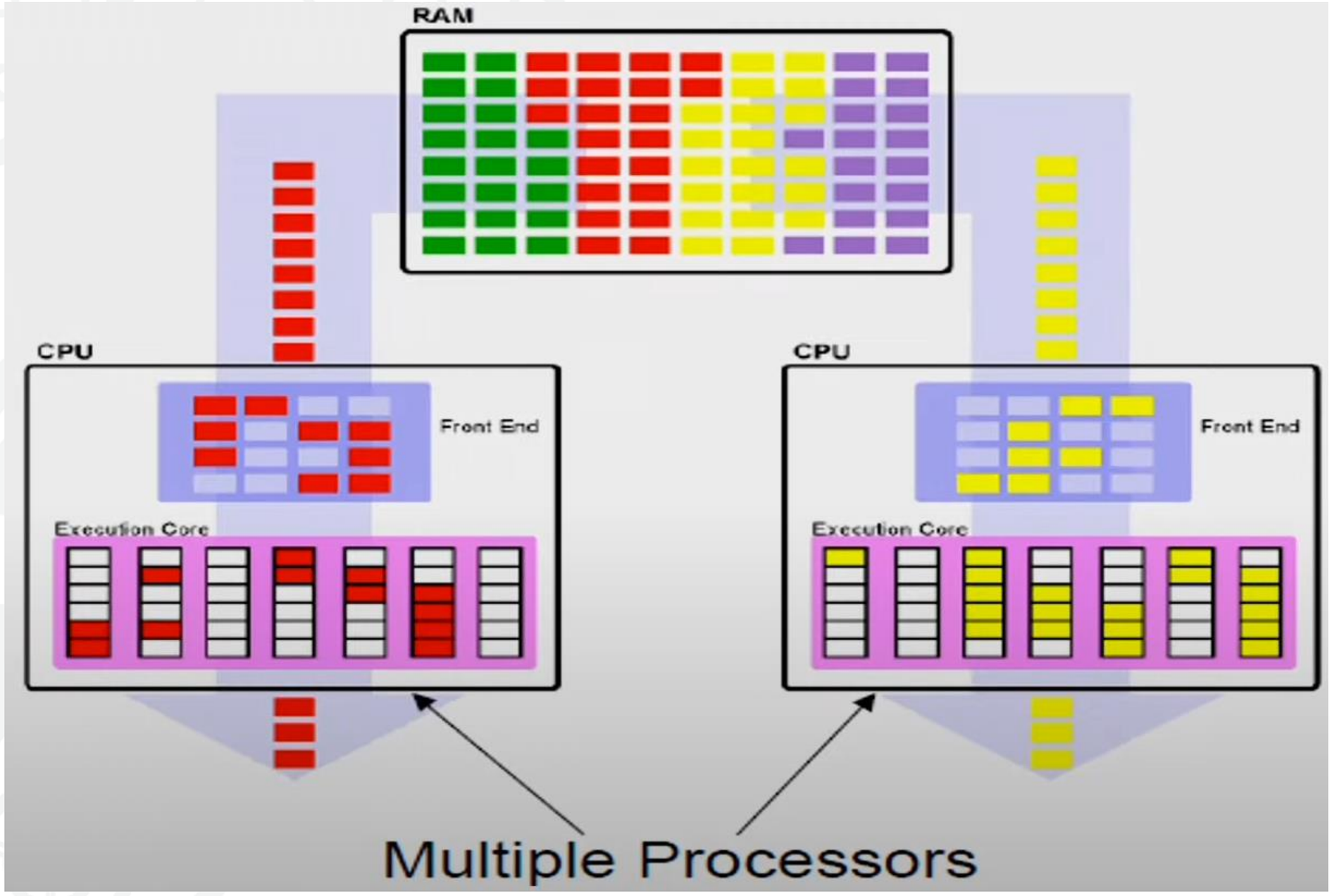
| Iteration number | Instructions | Issues at clock number | Executes at clock number | Read access at clock number | Write CDB at clock number | Commits at clock number | Comment |
|---|---|---|---|---|---|---|---|
| 1 | LD R2,0(R1) | 1 | 2 | 3 | 4 | 5 | First issue |
| 1 | DADDIU R2,R2,#1 | 1 | 5 | | 6 | 7 | Wait for LW |
| 1 | SD R2,0(R1) | 2 | 3 | | | 7 | Wait for DADDIU |
| 1 | DADDIU R1,R1,#8 | 2 | 3 | | 4 | 8 | Commit in order |
| 1 | BNE R2,R3,LOOP | 3 | 7 | | | 8 | Wait for DADDIU |
| 2 | LD R2,0(R1) | 4 | 5 | 6 | 7 | 9 | No execute delay |
| 2 | DADDIU R2,R2,#1 | 4 | 8 | | 9 | 10 | Wait for LW |
| 2 | SD R2,0(R1) | 5 | 6 | | | 10 | Wait for DADDIU |
| 2 | DADDIU R1,R1,#8 | 5 | 6 | | 7 | 11 | Commit in order |
| 2 | BNE R2,R3,LOOP | 6 | 10 | | | 11 | Wait for DADDIU |
| 3 | LD R2,0(R1) | 7 | 8 | 9 | 10 | 12 | Earliest possible |
| 3 | DADDIU R2,R2,#1 | 7 | 11 | | 12 | 13 | Wait for LW |
| 3 | SD R2,0(R1) | 8 | 9 | | | 13 | Wait for DADDIU |
| 3 | DADDIU R1,R1,#8 | 8 | 9 | | 10 | 14 | Executes earlier |
| 3 | BNE R2,R3,LOOP | 9 | 13 | | | 14 | Wait for DADDIU |

**The time of issue, execution, and writing result for a dual-issue version of our pipeline *with* speculation.**

# Superscalar Processor



# Symmetric Multiprocessor

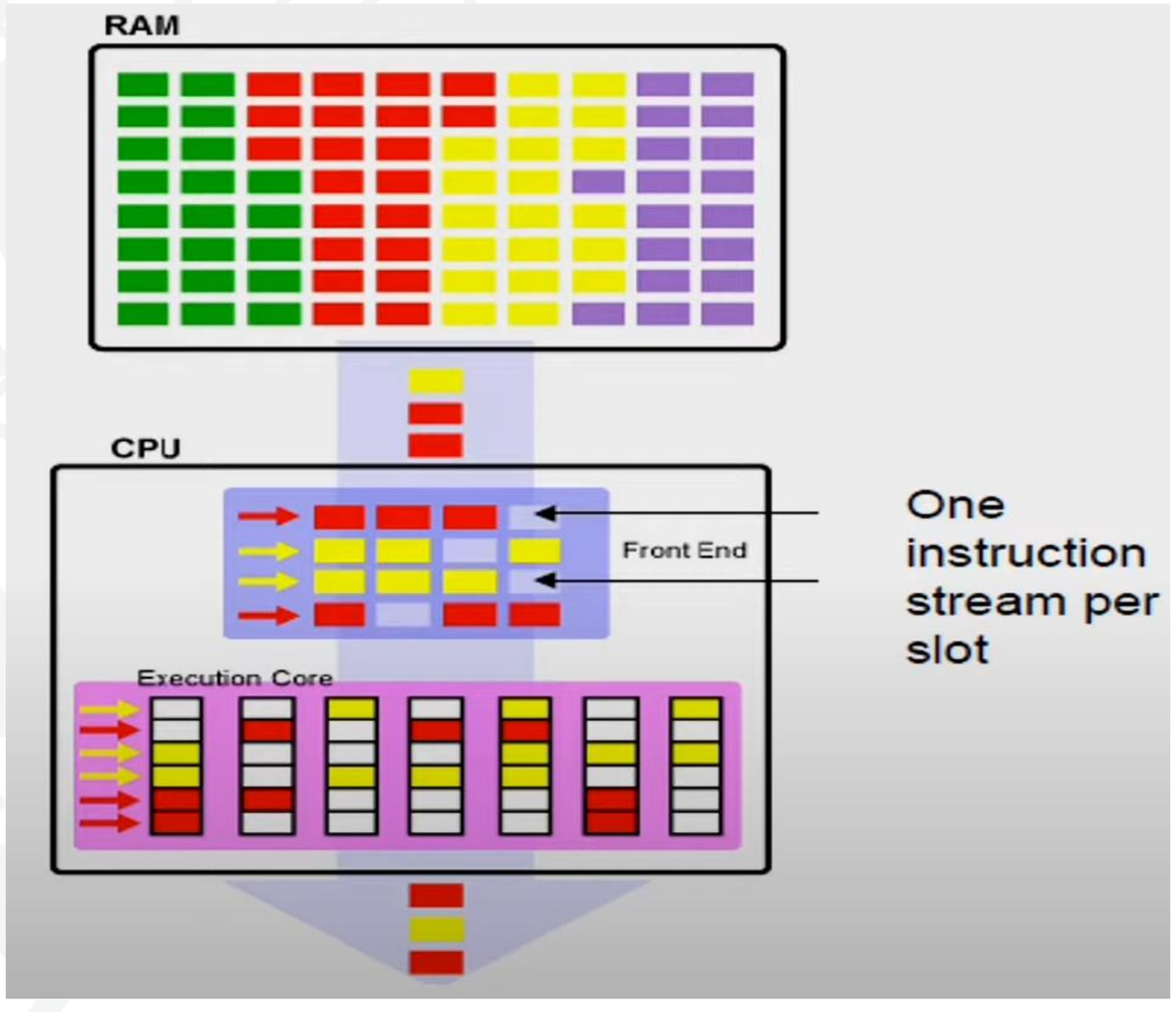


# End of Exploiting ILP?

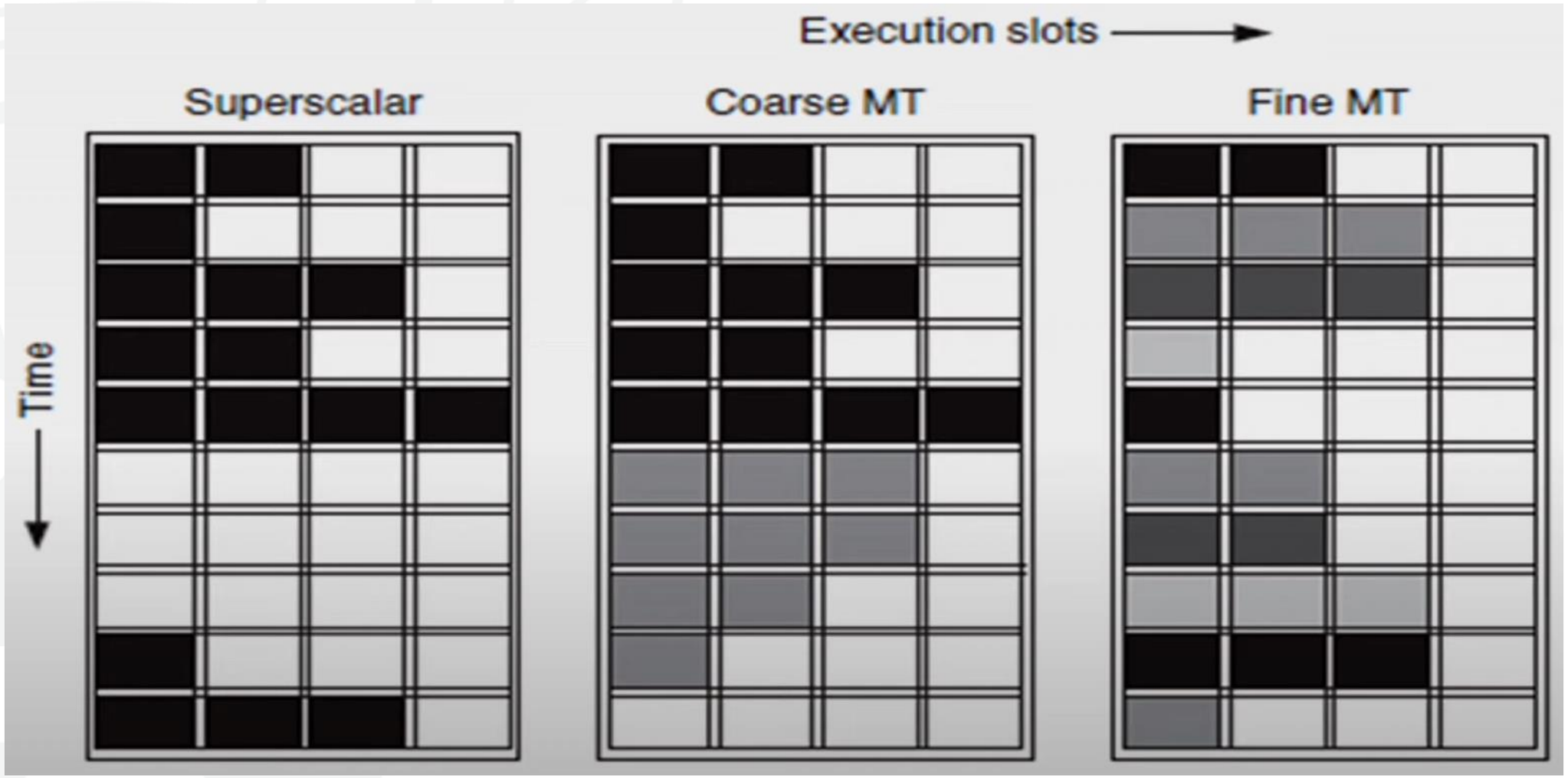
- We have tried Tomasulo's superscalar approach with ROB, speculation with aggressive branch prediction and multi issue
- At very high issue rates cache misses that go to L2 and off chip can not be hidden by ILP
- How to cover such long memory stalls?
- **Multithreading** allows multiple treads to share the functional units of a single processor in an overlapping fashion

# Multithreading

## Multithreading

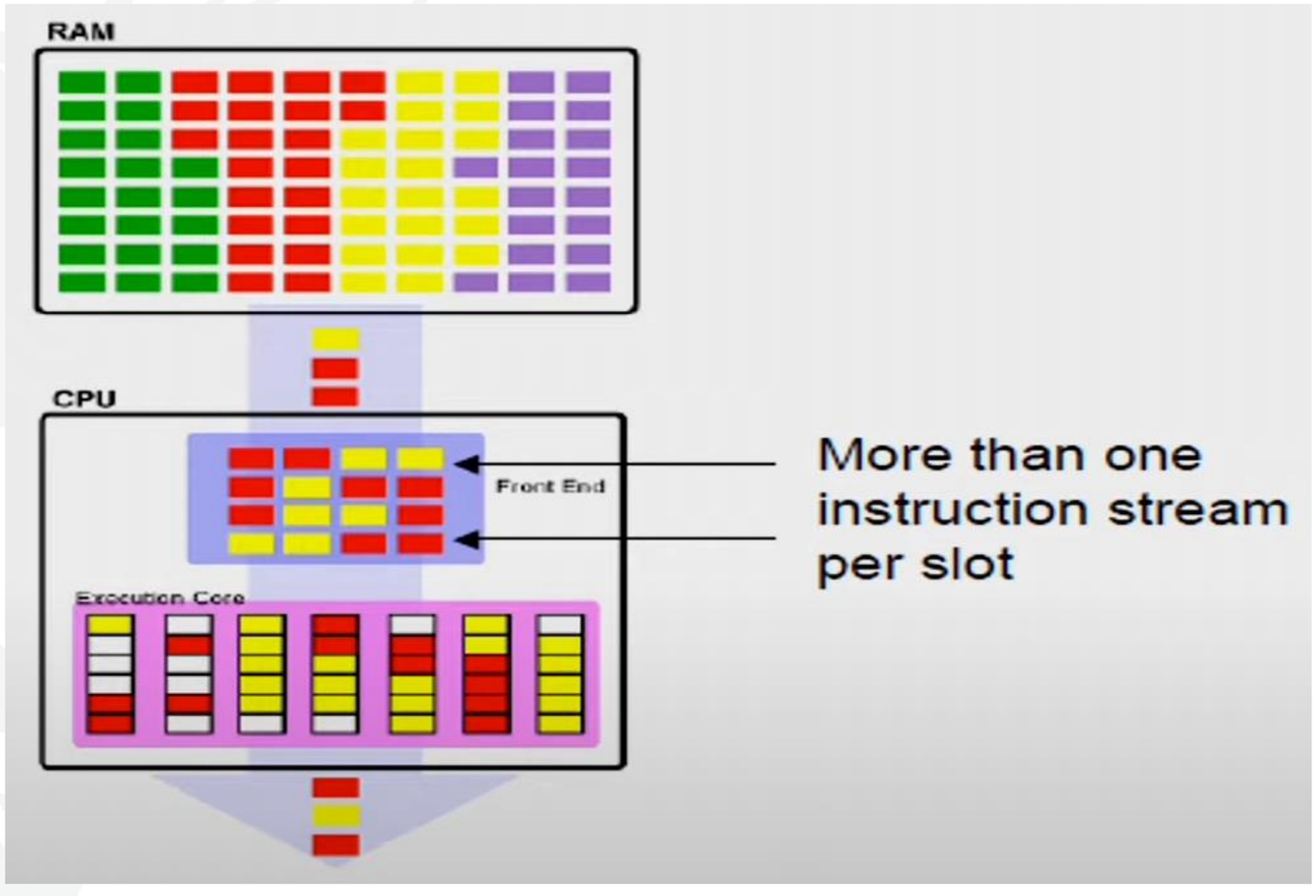


# Multithreading

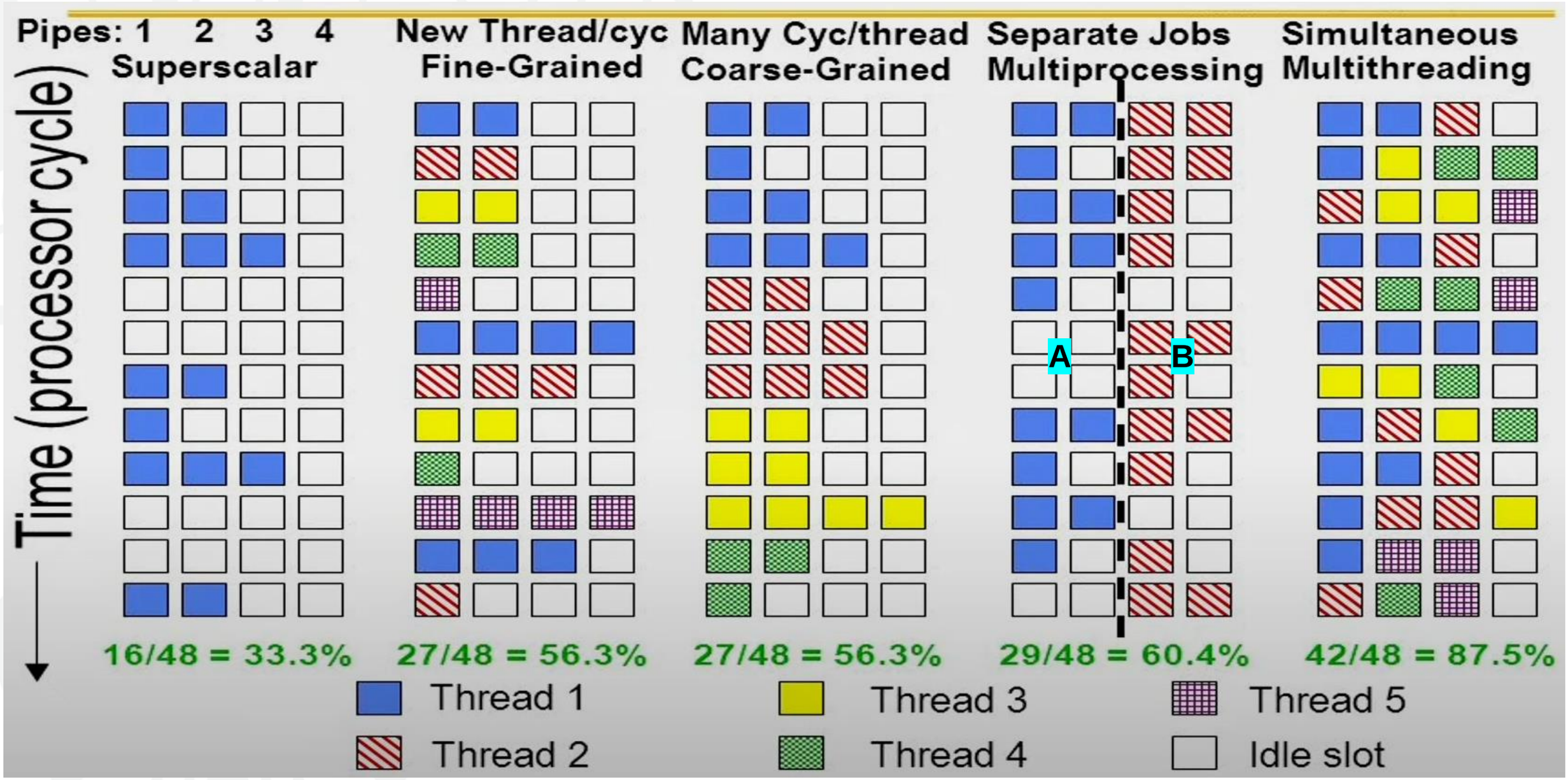


# Hyperthreading / SMT

## Hyperthreading



# Comparison of Resource Utilization



# Summary

# Suggested Readings

Computer Architecture: A Quantitative Approach (5th Edition)
by John Hennessy and David A Patterson:

- pp. 192 - 202: Sections 3.7 and 3.8